§3.1.

P1.
$$\begin{cases} \dfrac{dx_1}{dt} = x_2 \\ \dfrac{dx_2}{dt} = x_3 \\ \dfrac{dx_3}{dt} = -x_2^2 \end{cases}$$

$$\left(x_1(t) = y,\ x_2(t) = \frac{dy}{dt},\ x_3(t) = \frac{d^2y}{dt^2} \right)$$

P2.
$$\begin{cases} \dfrac{dx_1}{dt} = x_2 \\ \dfrac{dx_2}{dt} = x_3 \\ \dfrac{dx_3}{dt} = e^t - \cos x_1 \end{cases}$$

$$\left(x_1(t) = y,\ x_2(t) = \frac{dy}{dt},\ x_3(t) = \frac{d^2y}{dt^2} \right.$$

P3.
$$\begin{cases} \dfrac{dx_1}{dt} = x_2 \\ \dfrac{dx_2}{dt} = x_3 \\ \dfrac{dx_3}{dt} = x_4 \\ \dfrac{dx_4}{dt} = 1 - x_3 \end{cases}$$

$$\left(x_1(t) = y,\ x_2(t) = \frac{dy}{dt} \right.$$
$$\left. x_3(t) = \frac{d^2y}{dt^2},\ x_4(t) = \frac{d^3y}{dt^3} \right)$$

P4.
$$\begin{cases} \dfrac{dx_1}{dt} = x_2 \\ \dfrac{dx_2}{dt} = -2x_1 - 3x_4 \\ \dfrac{dx_3}{dt} = x_4 \\ \dfrac{dx_4}{dt} = -3x_2 - 2x_3 \end{cases}$$

P7. $\dot{x} = \begin{bmatrix} 5 & 5 \\ -1 & 7 \end{bmatrix} x, \quad x(3) = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$

P10. $x + y = \begin{pmatrix} 0 \\ 3 \\ 6 \end{pmatrix}$

$3x - 2y = \begin{pmatrix} 5 \\ 9 \\ -2 \end{pmatrix}$

P13. (a) $Ax = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$

(b) $Ax = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$

(c) $Ax = \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix}$

P14. Solution:

$$\begin{pmatrix} 3 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ -2 \\ -2 \end{pmatrix}$$
$$= \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 2\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$
$$\Rightarrow A\begin{pmatrix} 3 \\ -1 \\ -1 \end{pmatrix} = A\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 2A\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$
$$= \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$
$$= \begin{pmatrix} 5 \\ 6 \\ 12 \end{pmatrix}$$